What could possibly make the ruler shrink, when its orientation changes? Gravity, of course. In the flat space of the right half of Figure 11.1 there resides a gravitational field that controls the sizes of fundamental particles, atomic nuclei, atoms, molecules, everything, and forces them all to shrink when laid out radially. The amount of shrinkage is great near a black hole, and smaller farther away, because the shrinkage-controlling gravitational field is generated by the hole, and its influence. declines with distance.

The shrinkage-controlling gravitational field has other effects. When a photon or any other particle flies past the hole, this field pulls on it and deflects its trajectory. The trajectory is bent around the hole; it is curved, as measured in the hole's true, flat spacetime geometry. However, people like Einstein, who take seriously the measurements of their rubbery rulers and clocks, regard the photon as moving along a straight line through curved spacetime.

What is the real, genuine truth? Is spacetime really flat, as the above paragraphs suggest, or is it really curved? To a physicist like me this is an uninteresting question because it has no physical consequences. Both viewpoints, curved spacetime and flat, give precisely the same predictions for any measurements performed with perfect rulers and clocks, and also (it turns out) the same predictions for any measurements performed with any kind of physical apparatus whatsoever. For example, both viewpoints agree that the radial distance between the horizon and the circle in Figure 11.1, *as measured by a perfect ruler,* is 37 kilometers. They disagree as to whether that measured distance is the "real" distance, but such a disagreement is a matter of philosophy, not physics. Since the two viewpoints agree on the results of all experiments, they are physically equivalent. Which viewpoint tells the "real truth" is irrelevant for experiments; it is a matter for philosophers to debate, not physicists. Moreover, physicists can *and do* use the two viewpoints interchangeably when trying to deduce the predictions of general relativity.

The mental processes by which a theoretical physicist works are beautifully described by Thomas Kuhn's concept of a *paradigm* Kuhn, who received his Ph.D. in physics from Harvard in 1949 and then became an eminent historian and philosopher of science, introduced the concept of a